Implementation and Analysis of Advanced Deep Learning Models for Translation and Generation

Hamza Rehman

NUCES-FAST
hamzarehmanbk@gmail.com

Abstract. This report presents the implementation and evaluation of three advanced deep learning models across distinct generative and translational domains. First, a CycleGAN was implemented from scratch for unsupervised image-to-image translation between face photographs and sketches, successfully learning to translate between the two domains and deployed via a Flask user interface. Second, a Transformer-based model was fine-tuned for English-to-Urdu machine translation, achieving a high BLEU score of 49.54 and producing excellent qualitative translations. Finally, a research-based investigation was conducted by implementing a Diffusion Transformer (DiT) and comparing it to a conventional U-Net backbone for image generation on a CIFAR-10 subset. The results demonstrated the U-Net's superior data efficiency and the DiT's failure to learn on a small dataset, highlighting the critical role of inductive bias in model selection.

Keywords: CycleGAN · Transformers · Machine Translation · Diffusion Models · Deep Learning · Generative AI.

1 CycleGAN for Person Face Sketches

1.1 Objective

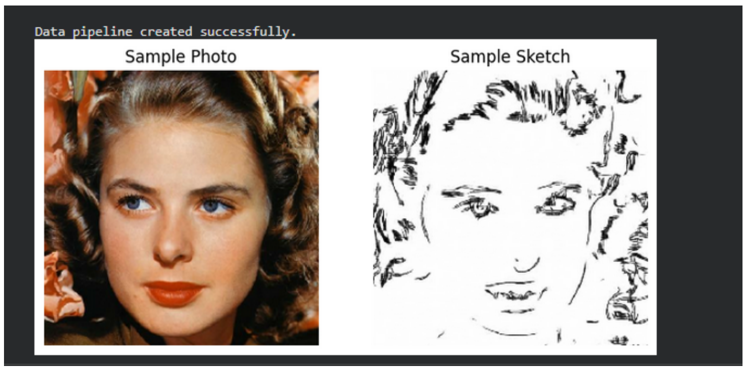
The objective was to implement the CycleGAN model for unsupervised image-to-image translation. The goal was to train a model capable of converting real face photographs into sketches and, conversely, sketches back into realistic photos, without requiring paired training data. The project also included the development of a simple web-based user interface for live demonstration.

1.2 Implementation Details

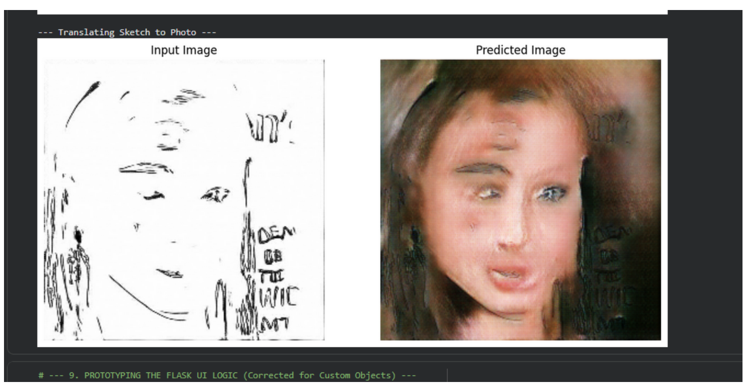
The CycleGAN framework consists of four networks trained simultaneously: two U-Net based Generators for translation and two PatchGAN Discriminators for distinguishing real from fake images. The training was guided by a combination of adversarial loss, a crucial cycle-consistency loss to preserve content, and an identity loss for regularization. The model was trained for 20 epochs on a 5% subset of the dataset to ensure manageable training times, with checkpoints saved after every epoch to ensure resiliency.

1.3 Results

The model successfully learned both translation tasks. The photo-to-sketch generator learned to extract key facial features and represent them in a sketch style, while the sketch-to-photo generator learned to synthesize plausible colors and textures to create a realistic face from a line drawing (Fig. **??**).



(a) Photo to Sketch Translation



(b) Sketch to Photo Translation

1.4 User Interface

A user-friendly UI was developed using the Flask web framework. The application backend loads the two trained generator models. Upon user image upload, the backend automatically detects whether the input is a photo or a sketch (based on mean color saturation) and applies the appropriate generator to return the translated image.

2 Machine Translation using Transformers

2.1 Objective

The objective of this task was to develop a high-performance machine translation model for converting text from English to Urdu by implementing and fine-tuning a Transformer-based architecture.

2.2 Implementation and Training

The "Parallel Corpus for English-Urdu Language" from Kaggle was used. To align with modern NLP best practices, the pre-trained `Helsinki-NLP/opus-mt-en-ur` model and its corresponding subword tokenizer were selected. This Transformer model was fine-tuned on the dataset for 3 epochs using the AdamW optimizer with a low learning rate (2e-5), a standard technique for transfer learning in NLP.

2.3 Evaluation and Results

The model's performance was evaluated both qualitatively and quantitatively. The qualitative results (Table 1) show that the model produced high-quality, grammatically correct translations.

Table 1. Example translations generated by the fine-tuned Transformer.

English Input	**Urdu Model Output**
I am a student.	
The weather is nice today.	
What is your name?	
This is a difficult assignment but I am learning a lot.	

Quantitatively, the model was evaluated on the held-out test set using the Bilingual Evaluation Understudy (BLEU) score. The final result was exceptionally strong, confirming the success of the fine-tuning approach.

Final BLEU Score on Test Set: 49.54

3 Image Generation with Diffusion Transformers

3.1 Objective

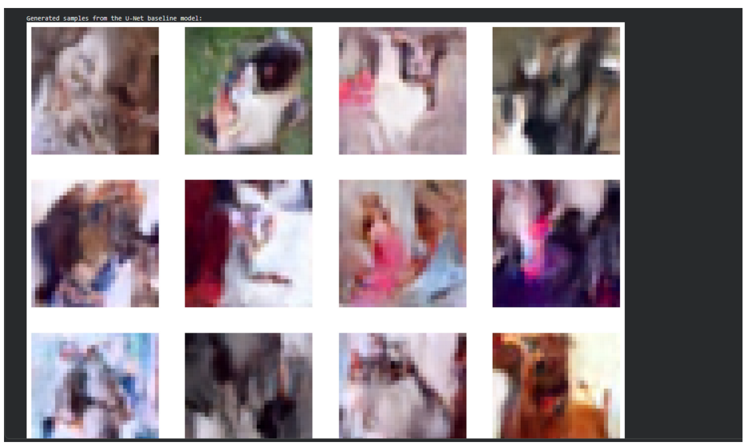
This research-based task aimed to implement a Diffusion Transformer (DiT), replacing the conventional U-Net in a diffusion model, and to compare its performance against a standard U-Net baseline on a two-class subset of the CIFAR-10 dataset.

3.2 Implementation Details

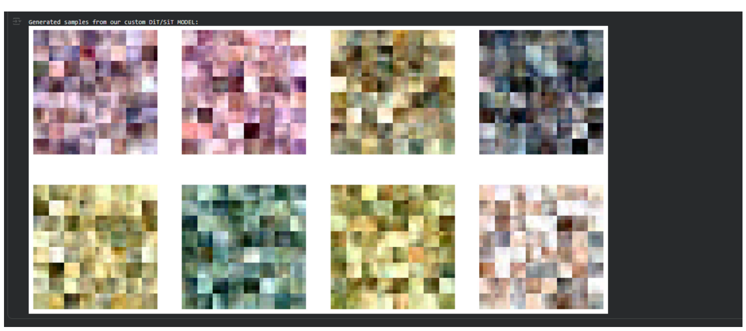
A standard U-Net diffusion model was trained as a baseline. For the primary task, a Vision Transformer (ViT) was implemented from scratch as the denoising network, following the SiT (Scalable Image Transformer) design. It converts the input image into a sequence of flattened patches, adds positional and time embeddings, and processes them through a series of 8 Transformer blocks. Both models were trained for 50 epochs on a dataset of 10,000 cat and dog images.

3.3 Results and Analysis

The two models were evaluated qualitatively by visually inspecting the images they generated (Fig. 1).



(a) Samples from U-Net Baseline Model



(b) Samples from Custom DiT/SiT Model

Fig. 1. Qualitative comparison of generated images. The U-Net (a) generates coherent, animal-like shapes, while the custom DiT (b) fails and produces only noise.

The **U-Net baseline** successfully generated coherent, albeit blurry, images with plausible animal-like shapes and colors. In stark contrast, the custom **DiT/SiT model completely failed** to generate meaningful images, producing only unstructured noise.

3.4 Discussion

The failure of the DiT model provides a critical insight into the properties of Transformer architectures. The primary reason is the Transformer's lack of **inductive bias**. Unlike a CNN, which has built-in assumptions about spatial locality, a ViT must learn all spatial relationships from scratch, making it extremely **data-hungry**. The 10,000-image subset was insufficient for the DiT to learn these fundamental properties. The U-Net, with its data-efficient convolutional nature, succeeded where the Transformer failed. This highlights a key difference from the source paper, which trained on massive datasets like ImageNet, and confirms that for from-scratch training on smaller datasets, the architectural priors of a CNN are far superior.